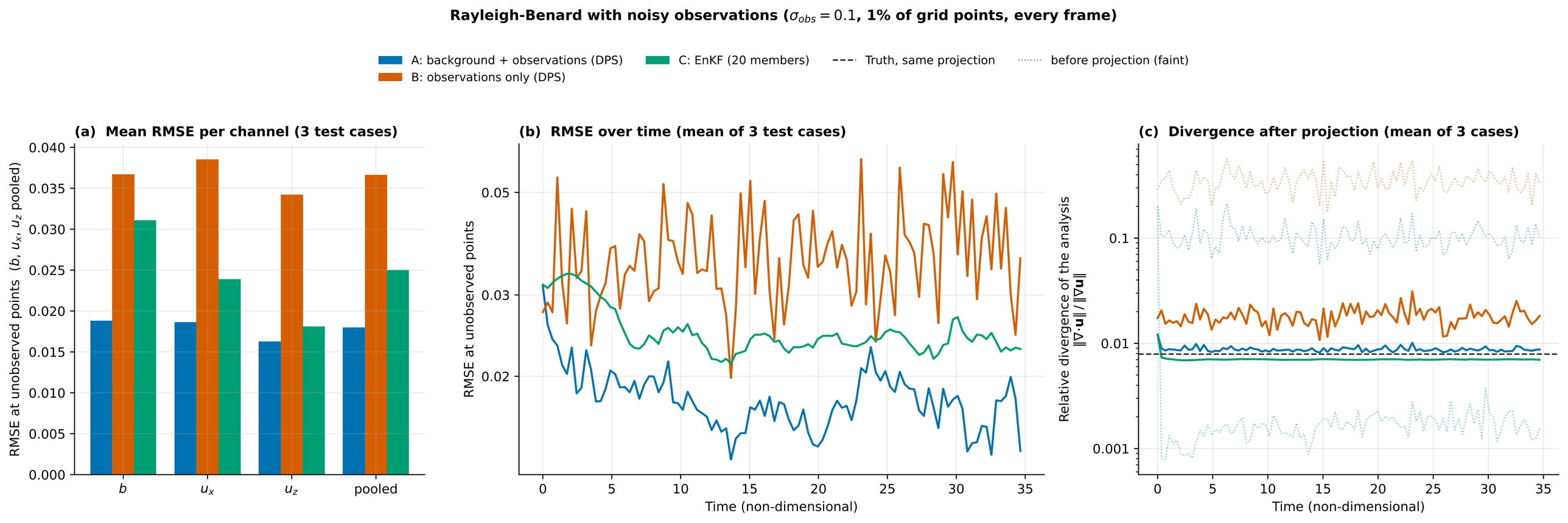
Rayleigh-Benard with noisy observations ($\sigma_{obs} = 0.1$, 1% of grid points, every frame)
A: background + observations (DPS)
B: observations only (DPS)
C: EnKF (20 members)
Truth, same projection
before projection (faint)
(a) Mean RMSE per channel (3 test cases)
RMSE at unobserved points (b, u_x, u_z pooled)
0.040
0.035
0.030
0.025
0.020
0.015
0.010
0.005
0.000
b
u_x
u_z
pooled
(b) RMSE over time (mean of 3 test cases)
RMSE at unobserved points
0.05
0.03
0.02
0
5
10
15
20
25
30
35
Time (non-dimensional)
(c) Divergence after projection (mean of 3 cases)
Relative divergence of the analysis
$\|\nabla\cdot\mathbf{u}\| / \|\nabla\mathbf{u}\|$
0.1
0.01
0.001
0
5
10
15
20
25
30
35
Time (non-dimensional)